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Vehicle headlamp lighting control module for adas and autonomous vehicles

Abstract

Vehicle headlamps are typically used to illuminate scenes for better vision. The same headlamps can be used as part of an active camera at discrete times during the camera's imaging periods. In embodiments, a lighting control module utilizes an electronic control module to control on/off states of the headlamps during a set of non-imaging periods of the camera system and a camera control configured to control multiple on/off states of the set of headlamps during a set of imaging periods of the camera system. Embodiments of lighting control modules can be used in Advanced Driver Assistance Systems (ADAS) and Autonomous Control Systems for commercial and passenger vehicles.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/967,365, filed on Oct. 17, 2022 and issued on Dec. 5, 2023 as U.S. Pat. No. 11,838,626, which is a continuation of U.S. patent application Ser. No. 17/127,461, filed on Dec. 18, 2020 and issued on Oct. 18, 2022 as U.S. Pat. No. 11,477,363, which is a continuation of U.S. patent application Ser. No. 16/537,305, filed on Aug. 9, 2019 and issued on Dec. 22, 2020 as U.S. Pat. No. 10,873,738, which is a division of U.S. patent application Ser. No. 16/167,196, filed on Oct. 22, 2018 and issued on Aug. 13, 2019 as U.S. Pat. No. 10,382,742, which is a continuation of U.S. patent application Ser. No. 15/853,222, filed on Dec. 22, 2017 and issued on May 21, 2019 as U.S. Pat. No. 10,298,908, which is a continuation of U.S. patent application Ser. No. 15/059,811, filed on Mar. 3, 2016 and issued on Jan. 9, 2018 as U.S. Pat. No. 9,866,816. All of the aforementioned applications are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) Embodiments relate generally to cameras and sensors that provide measurements of distances to points in an image of a scene. More particularly, embodiments relate to an active light source sensor system whereby images of scenes and objects are acquired within a broad range of ambient lighting conditions.

BACKGROUND

(2) Three dimensional photonic imaging systems, also referred to as three-dimensional (3D) cameras, are capable of providing distance measurements and photonic measurements for physical objects in a scene. Applications for such 3D cameras are industrial inspection, selective robot vision, 3D modeling, geographic surveying, and forensic analysis.

(3) 3D cameras can be implemented with a variety of technologies, with each combination of technologies presenting certain limitations that render the cameras ineffective in broad-use applications. Stereo vision 3D cameras implement two or more imaging arrays in a fixed, highly-calibrated configuration and utilize triangulation of common points within the fields of view to establish distances to each of the common points. Stereo vision systems suffer from image distance inaccuracies due to occlusion and parallax. Furthermore, distance accuracies suffer when the baseline distance between image arrays is small relative to the distances being measured. Lastly, stereo 3D cameras are expensive due to the need for multiple image arrays and the requirement for high precision for the baseline offset between the image arrays.

(4) Time-of-flight (TOF) systems utilize light sources, such as lasers, that are pulsed or modulated so they provide pulses of light for illuminating scenes in conjunction with an imaging system for measuring the amplitude and timing of the light reflected from the objects in the scene. Distances to points in the scene are determined using the known speed of light for all of the reflected signals. The imaging systems for TOF devices comprise a camera with a photodetector array, typically fabricated using CCD or CMOS technology, and a method for rapidly gating the collection times for the photodetector elements. Reflected light is captured by the photodetector elements during the specific gating cycles.

(5) Some TOF systems only utilize the timing between light pulses and gated photodetectors to

determine 3D object distances. Other TOF systems utilize the amount of received light during a gated capture cycle to establish object distances. The accuracy of these systems depends on the uniformity of incident light and the speed of the gating mechanism for the photodetectors.

(6) Utilizing gated photodetectors is an effective method to establish distances to objects in a scene. By precisely controlling the timing between incident light pulses and gated photodetectors the distances to objects in certain distance bands can be accurately determined. For establishing object distances for other distance bands, subsequent light and gated photodetector cycles are utilized while the stationary objects and stationary camera are maintained in their present configurations and orientations. Any movement of the camera and/or objects in the scene will result in distance measurement bands that are not registered with one another.

(7) A 3D camera described in U.S. Pat. No. 4,935,616 utilizes a modulated source and imaging system. A preferred embodiment of this system uses a CW laser and utilizes the phase difference between the incident and reflected signals to establish the distances to objects.

(8) Another 3D camera is described in U.S. Pat. No. 5,081,530. This system utilizes a pair of gates for each photodetector element. Distances to objects are determined from the ratio of differences between the sampled energy at the two gated elements.

(9) U.S. Pat. Nos. 7,362,419 and 7,755,743 each utilize modulated light intensity sources and phase ranges to detect phase shifts between emitted and detected signals. An embodiment of U.S. Pat. No. 8,159,598 utilizes modulated light intensity and phase shift detection for time of flight determination. Other embodiments of U.S. Pat. No. 8,159,598 utilize a high resolution color path with a low resolution distance path to determine 3D information for a detector or a group of detectors.

(10) U.S. Pat. No. 8,102,426 to Yahav describes 3D vision on a chip and utilizes an array of photodetector elements that are gated at operative times to establish object distances in a scene. Photodetector sites are utilized for either TOF distance measurement or for the determination of object color. Embodiments of Yahav describe utilizing groups or bands of photodetector elements to establish the various distance bands. Other embodiments of Yahav describe a single distance band for each capture cycle, with full scene distances established utilizing sequences of capture cycles. Although not specified in Yahav, the requirement for the embodiments is no movement of the camera and the objects in the scene throughout the sequence of capture cycles.

(11) For real-world applications like autonomous vehicle navigation, mobile mapping, agriculture, mining, and surveillance it is not practical to require little or no movement between a 3D camera and objects in a scene during a sequence of imaging cycles. Furthermore, most of the real-world situations occur in scenes that have widely-varying ambient light conditions. Geiger mode avalanche photo diodes are solid state photodetectors that are able to detect single photons. Such Geiger mode avalanche photo diodes are also referred to as single-photon avalanche diodes (SPADs). Arrays of SPADs can be used as a single detector element in an active sensing system, but camera/sensor systems based on SPAD arrays have at least two shortcomings due to ambient light. First, solar background light can hamper the ability to accurately determine depth. Second, ambient light impacts the reflectivity precision because of challenges differentiating between reflected light and ambient light. It is desirable to have a 3D camera/sensor with one or more SPAD arrays that can address these shortcomings.

SUMMARY

(12) In embodiments, an active sensor system is configured to generate a lighting-invariant image of a scene utilizing at least one emitter configured to emit a set of active light pulses toward the scene. A focal plane array of Geiger mode avalanche photo diode detectors is configured to receive light for a field of view that includes at least a portion of the scene, wherein each detector is biased to operate as a single-photon avalanche diode (SPAD) detector in an array of SPAD detectors. Control circuitry is operably coupled to the at least one emitter and the array of SPAD detectors and is configured to emit the set of light pulses and to capture a set of intensity values for at least three

successive distance range bands and store the set of captured intensity values in a set of frame buffers. A processing system is operably coupled to the control circuitry and the set frame buffers to generate the lighting-invariant image of the scene. In embodiments, the processing system is configured to analyze the at least three frames and determine a minimum intensity value due to ambient light, a maximum intensity value, and a frame of the at least three successive frames at which the maximum intensity value occurs, determine a depth based on the frame at which the maximum intensity value occurs, determine a reflectivity value based on the difference between the maximum intensity value and the minimum intensity value, and generate the lighting invariant depth-map of the scene based on the depths and the reflectivity values.

(13) In embodiments, the incident light is full spectrum visible light in the energy band from roughly 400 nanometers to 700 nanometers. The photodetector sites are sensitive to radiation in this wavelength band. In embodiments the photodetectors utilize a bandpass filter to reduce or eliminate radiation outside the desired energy band. The bandpass filter(s) can be applied as a global array filter in the case of an IR-eliminating filter and can be applied as an individual filter for each photodetector in the case of a Bayer pattern for establishing RGB elements in the array.

(14) In some embodiments, a photodetector element utilizes a photodiode coupled to a photodetector integration element whereby the current from the photodiode produces a charge that is collected or integrated during the gating cycle of the photodetector. The photodetector integration stage is emptied by rapidly transferring the integrated charge to the next processing element in the system, thus allowing the photodetector stage to begin the integration for the subsequent photodetector integration cycle.

(15) In embodiments, each photodetector site in the array is connected to a dedicated charge transfer stage wherein each of the K stages facilitates the rapid transfer of charges from the photodetector site. Utilizing K charge transfer stages per photodetector allows for up to K gated emitter/detector cycles per imaging cycle.

(16) In embodiments, the detector array and charge transfer sites are fabricated together on a focal plane array along with gating circuitry and charge transfer control circuitry. The number of photodetector sites will be sufficiently large and the focal plane array interface to the downstream camera circuitry will be of a relatively lower throughput rate than that of the throughput of the higher-speed charge transfer array.

(17) In embodiments, the detector array is fabricated on a focal plane array along with gating circuitry. The signal interface for the focal plane array is sufficiently fast that the integrated charges can be transferred from the integration sites directly to a 4D frame buffer without the need for a charge transfer array.

(18) In some embodiments, the 4D camera light source comprises one or more elements like LEDs that provide uniform intensity throughout the desired frequency range. In other embodiments, the light source is a combination of light elements like LEDs wherein the frequency responses on the separate light elements combine to form an output signal that is uniform throughout the desired frequency range.

(19) In embodiments, the camera detects environmental conditions that attenuate emitted and reflected signals and utilizes detected information from environmental and ambient signals to establish non-attenuated signal strength, object distances and object color.

(20) In embodiments, the light intensity is non-uniform throughout the frequency range. The non-uniform light is characterized to establish parameters for use in color correction applied during image post-processing. In embodiments the light intensity is spatially non-uniform throughout the field of view. The non-uniform spatial intensity is mapped to allow for scene intensity and color adjustments during image post-processing.

(21) In some embodiments, the light energy is emitted and received as common laser wavelengths of 650 nm, 905 nm or 1550 nm. In some embodiments the light energy can be in the wavelength ranges of ultraviolet (UV)—100-400 nm, visible—400-700 nm, near infrared (NIR)—700-1400

nm, infrared (IR)—1400-8000 nm, long-wavelength IR (LWIR)—8 μ m-15 μ m, far IR (FIR)—15 μ m-1000 μ m, or terahertz—0.1 mm-1 mm.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an external view of a 4D camera configuration.
- (2) FIG. 2 illustrates aspects of a 4D camera that define a system's field of view.
- (3) FIG. 3 illustrates a functional block diagram of the main electrical and optical elements of a system.
- (4) FIG. 4 illustrates an electrical timing diagram showing multiple 4D image capture and processing cycles.
- (5) FIG. 5 illustrates an electrical functional block diagram for a detector array where each photodetector pixel consists of a single photodetector, an associated integration stage, and K charge transfer stages.
- (6) FIG. 6 illustrates an electrical timing diagram for eleven stages of emitter/detector events that constitute data collected for a single 4D frame buffer.
- (7) FIG. 7 illustrates TOF ranges for a typical 16-stage detector pixel and shows the associated collected intensity values for all integration stages.
- (8) FIG. 8 illustrates imaging of an object with fog obscuring an optical path.
- (9) FIG. 9 illustrates TOF ranges and intensity values for imaging with fog obscuring an optical path.
- (10) FIG. 10 illustrates an electrical timing diagram for ten stages of detector events for a single 4D frame buffer wherein the final stage is expanded in time to integrate more color information.
- (11) FIG. 11 illustrates TOF ranges for a ten-stage detector pixel and shows associated collected intensity values for all integration stages.
- (12) FIG. 12a illustrates an optical timing diagram for a 50 nSec emitter pulse.
- (13) FIG. 12b illustrates an optical timing diagram for cumulative intensity of the same 50 nSec emitter pulse.
- (14) FIG. 13a illustrates a spectral output for an ideal visible spectrum emitter.
- (15) FIG. 13b illustrates a spectral output for a typical white LED emitter.
- (16) FIG. 14a illustrates a spectral output for red LED and green LED emitters along with white LED emitters.
- (17) FIG. 14b illustrates a cumulative spectral output for the combined signals generated with white, green, and red LEDs.
- (18) FIG. 15 shows the spectral output for blue, red, and green emitters.
- (19) FIG. 16 illustrates an automotive configuration where emitted light is generated by vehicle headlamps and detector optical and electrical components are housed inside a vehicle separated from emitters.
- (20) FIG. 17 illustrates a headlamp control timing in an automotive configuration whereby 4D camera circuitry assumes control of the headlamps.
- (21) FIG. 18 illustrates an environment for the collection of image data for use in angular intensity profile analysis for an object.
- (22) FIG. 19 illustrates the flow diagram for an algorithm that utilizes angular intensity profile analysis to classify or identify an object.
- (23) FIG. 20 illustrates an automotive configuration where cameras are used to provide information in low visibility conditions.

DETAILED DESCRIPTION OF THE DRAWINGS

- (24) FIG. 1 shows a mechanical configuration for a 4D camera 10. The lens 20 comprises one or

more transparent optical elements that guide and focus light from the field of view (FOV) onto a detector array. Bandpass filters are typically included as part of the lens **20** assembly to filter out unwanted frequency components. For example, a 4D camera **10** that utilizes 905 nm lasers for emitters may include a narrow bandpass filter at 905 nm in order to exclude visible spectrum and other IR frequencies from energizing the detector elements. Emitters **30** provide incident light that is distributed over an area that includes the FOV of the detector array and associated lens **10**. Light originating from emitters **30** is “active” light in that it can be distinguished from “passive” ambient light. Individual emitters **30** may be lensed to properly distribute light throughout the FOV. All emitters **30** may generate the same frequencies as each other or separate emitters **30** may generate different bands of a desired spectral output of a device **10**. Pulsing of the emitters **30** is precisely controlled by electronics on board the 4D camera **10**. Mechanical housing **40** for the camera **10** will typically have device mounting features to allow for precise orientation of a camera **10** to a device or vehicle to which it is attached. Power is supplied to 4D camera **10** via a power connection **45**. Information is transferred to/from camera **10** either via an integrated power/communication cable **45**, via a separate I/O cable, or via a wireless interface.

(25) FIG. 2 shows an optical configuration of an embodiment for a detector array **50**. A lens **52** or other optics determines a field of view **56** of a detector array **50**. The edges of a detector array **50** field of view **56** are defined by angular extents from the normal vector of the device. This same normal vector will serve as a reference vector for subsequent angular measurements for detected objects. A surface **54** shown is a portrayal of a detector array **50** field of view at a constant distance from an array **50**. In various embodiments, each detector element **58** in an array **50** may be associated with a different angle **60** within a field of view **56**. An individual detector's **58** field of view **62** will be a subset of a detector array's **50** field of view **56**. Said another way, a detector array **50** field of view **56** is the summation of all of the individual fields of view **62**.

(26) For a detector array **50** with an in-focus lens **52** the individual fields of view **62** corresponding to each detector **58** should perfectly align with the fields of view for neighboring detectors. In practice, a lens **52** will almost never be perfectly in focus. Thus, the fields of view **62** of each detector **58** in a lensed system may typically overlap, though the field of view of each detector **58** is different from that of any other detector **58** in the detector array **50**. Detector arrays **50** may not have optimal density in their configuration due to semiconductor layout limitations, substrate heat considerations, electrical crosstalk avoidance, or other layout, manufacturing, or yield constraints. As such, sparse detector arrays **50** may experience loss in photon detector efficiency within the device field of view **56** due to reflected photons contacting the unutilized spaces between successive detector elements **58**.

(27) For non-lensed systems the field of view **62** of each detector **58** can be determined by a diffraction grating, an interferometer, a waveguide, a 2D mask, a 3D mask, or a variety of other aperture configurations designed to allow light within a specific field of view. These individual detector apertures will typically have overlapping fields of view **62** within the device field of view **56**.

(28) An element of various embodiments is the determination of an angle **60** for each detector **58**. FIG. 2 shows a detector array **50** with a single lens **52**. Another embodiment utilizes micro lenses at each detector element **58** wherein the individual micro lenses are configured to transmit reflected light at various angles throughout the device's field of view **56**. Another embodiment utilizes detector elements **58** with waveguides at various angles throughout the device's field of view. Other embodiments utilize detector elements with apertures created from interferometers, diffraction gratings, 2D masks, 3D masks, or other aperture-forming structures to create waveguide properties at various angles throughout the device's field of view. For a single-lens **52** system like that of FIG. 2 with a lens **52** configured to transmit in-focus light to the array **50**, the individual fields of view **62** are essentially adjacent to the fields of view of the neighboring detectors **58** in the array **50**. Out-of-focus lenses **52** will produce overlapping fields of view **62** for individual detectors **58**.

Waveguide and aperture detectors will likely produce overlapping fields of view **62** for individual detectors **58**. Micro lenses will also likely produce overlapping fields of view **62** for the individual detectors **58**. All of these overlapping field-of-view embodiments produce reliable results according to the specifications herein. The features of the optical detection system of these embodiments are that multiple detector elements comprise a device's field of view **56**, and every element in the detector array is defined by the angle **60** that determines the detector **58** field of view **62**.

(29) Variations will occur in the fabrication of detector arrays **50** used in 4D cameras. In single-lens **52** detector array devices like that shown in FIG. 2, miniscule differences in the alignment of the array **50** and the lens **52** can cause differences in the detector angles between separate devices. Because of the minor fabrication and assembly differences between devices, each device may undergo a post-production characterization process. The characterization process defines the central angle **60** for each as-constructed detector element. In various embodiments, characterization data from this process is stored in non-volatile memory or in a configuration file for every device. Waveguide, micro lens and aperture devices may require similar characterization to establish angles **60** for each detector element.

(30) Due to the importance of accurate determination of the optical path, in situ calibration may be desirable for devices according to various embodiments. As an example, a 4D camera device according to an embodiment may be used as a sensor in an autonomous vehicle. In order to protect the device it may be mounted inside a passenger vehicle affixed to the windshield behind the rear-view mirror. Since the device is facing in front of a vehicle, emitted light and reflected light will pass through the windshield on its way to and from external objects. Both components of light will undergo distortion when passing through the windshield due to reflection, refraction, and attenuation. In situ calibration for this autonomous vehicle 4D camera may include the device emitting pre-determined calibration patterns and measuring the intensity, location, and angle of the reflected signals. Device characterization parameters would be updated to account for a modified optical path of the incident and/or reflected light based on a calibration.

(31) FIG. 3 shows a functional block diagram for an embodiment with electrical and optical elements for a 4D camera **70**. The photodetector array **72** is a configuration of M rows by N columns of elements that sense photons at certain frequencies and convert optical energy to electrical information. An image capture cycle for the 4D camera **70** will consist of K successive emitter/detector cycles produced in a rapid sequence, with the relative timing varied for each of the K successive cycles to account for different TOFs and thus different distance range bands for the capture cycles. The information from the photodetector array **72** is stored in the 4D frame buffer memory **74**. There are K frame buffers in the 4D frame buffer memory **74**, with each of the K frame buffers corresponding to detector information collected for the M×N detectors **72** for each emitter/detector cycle. Information from the photodetectors **72** is typically collected and integrated in analog signal form. A/D converters **76** translate the detector **72** information to digital format prior to storage in the digital-information 4D frame buffers **74**. Each of the K 4D frame buffers **74** will have M×N×L bits, where M is the number of rows in the detector array **72**, N is the number of columns in the detector array **72**, and L is the number of bits per pixel produced by each A/D converter **76**. The number of A/D converter elements **76** in embodiments can range from 1 to M×N. More A/D converter elements **76** allow for faster transfer of information from a photodetector array **72** to a 4D frame buffer **74**.

(32) In embodiments a photodetector array **72** is fabricated as a focal plane array that utilizes electrical connections **78** to interface with other camera **70** circuitry. This electrical interface **78** is typically of lower bandwidth than that required by the high-speed photodetection elements **72**. The charge transfer array **80** is a collection of fast analog storage elements that takes information from the photodetector array **72** at a rate sufficient to allow the photodetector elements **72** to rapidly process subsequent emitter/detector events. The size of the charge transfer array **80** is typically M×N×K analog storage elements where M is the number of rows in the detector array **72**, N is the

number of columns in the detector array **72**, and K is the number of emitter/detector cycles that constitute a 4D capture cycle for a single 4D camera **70** event.

(33) Information from a 4D frame buffer **74** is processed separately for color information and distance information. A controller **82** computes distance values from the TOF algorithm for each of the $M \times N$ pixels and stores the distance information in the depth map **84** memory. In embodiments a photodetector array **72** is fabricated with a color filter pattern like a Bayer pattern or some other red-green-blue (RGB) configuration. Each color from a detector filter pattern will require a corresponding color plane **86** in device **70** memory. FIG. **3** shows color planes 0 through 2 **86** corresponding to, for example, red, green and blue planes of a detector array with a Bayer filter pattern. The number of color planes **86** can be one for monochrome, grey scale, or single-frequency applications and for IR and NIR applications. The number of color planes **86** is denoted as C and corresponds to the number of different bandpass filters utilized in the optical path of the detector array **72** or corresponds to the number of separate emitter frequencies that are utilized in a single-detector filtered, multi-emitter-wavelength configuration. Each of the C color planes **86** contains $M \times N$ elements, each with L bits of information. For embodiments with multi-colored detector filters a controller **82** may perform demosaicing on each of the C color planes **86** to create a dense color map from the sparse maps produced by the detector filter pattern.

(34) A controller **82** will assemble separate color planes **86** into an output image format and store a resulting file in device memory **88**. An output file may be in a format such as TIFF, JPEG, BMP or any other industry-standard or other proprietary format. Depth map **84** information for an image may be stored in the image file or may be produced in a separate file that is associated with the image file. After completion of the creation of the output file(s) the controller **82** transmits information via the I/O **90** interface to an upstream application or device. A controller **82** configures all of the sequencing control information for the emitters **92**, the photodetector **72** integration, the 4D frame buffer **74** transformation to color **86** and depth **84** information, and device **70** communication to other devices. A controller **82** can be a single CPU element or can be a collection of microcontrollers and/or graphics processing units (GPUs) that carry out the various control functions for the device **70**.

(35) FIG. **4** shows electrical signal timing for multiple image capture cycles. During the first sequence of emitter drive pulses **100** there are K pulses that energize the camera emitters. At select times following the start of the emitter drive pulses, K detector integration **102** cycles are performed. Upon completion of the last detector integration cycle **102** the information is transferred from a detector array or charge transfer array to a 4D frame buffer **104**. The time to transfer the $M \times N \times K$ intensity values from a detector array or charge transfer array to a 4D frame buffer is typically much longer than the time for emitter/detector cycles **100**, **102**.

(36) Upon completion of the filling of a 4D frame buffer a camera controller will create an $M \times N$ depth map **106** and will create the color plane(s) **108**. In embodiments where a camera utilizes multiple color planes produced by multiple color filters on a detector array a controller performs demosaicing for each of the sparse color planes to produce $M \times N$ color values for each color plane. A controller creates an output file for the present color image and will format **110** the file for transmission to the upstream device or application.

(37) The frame rate of a 4D camera will typically be a function of the longest action in the processing sequence. For the FIG. **4** embodiment the filling of the 4D frame buffer **104** is the longest action due to typical bandwidth constraints of focal plane array architecture. When the 4D frame buffer is emptied a new emitter/detector cycle can begin. Emitter drive pulses **114** and detector integration cycles **116** for a second 4D camera frame cycle are shown commencing while the depth **106** and color plane(s) **108** are created for the previous camera cycle. This pipelining of computation cycles is a technique known in the art for increasing the throughput of a normally serial sequence of computing events.

(38) FIG. **5** shows a functional electrical diagram for detector array **120** elements. A detector array

120 consists of **M** rows and **N** columns of elements. The functionality of a single detector element **122** is shown at the top of FIG. 5. A photodetector **124** is shown in a reverse-bias configuration. Incident photons **126** at the photodetector **124** cause a current to flow when the integration gate **128** is in the gated ON position. This gating switch allows current to flow to the integration stage **130** only during times when the gate **128** is closed. Current flowing to the integrator stage **130** allows charge to collect at the capacitor **132** via the amplifier **134**. It should be noted that the capacitor **123** is a functional element that represents any device or feature that collects charge due to a current flow from a photodetector **124**. Those persons skilled in the art can produce replacements for the gate switch **128**, integrator **134**, and capacitor **132** with equal functional components while conforming to the elements of embodiments.

(39) During a detector **122** integration cycle the intensity of the charge collected at the capacitor **132** is proportional to the number of incident photons **126** present during the gating time of the integrator **130**. During photodetector **130** integration the charge transfer switch **136** remains in the open position. Upon completion of an integration cycle the integration switch **128** is opened and collected charge remains at the integrator **130** stage. During the start of the charge transfer cycle charge is migrated from the integration capacitor **132** to the charge transfer stage 0 **138** capacitor **140** by closing the charge transfer stage 0 **138** gate switch **136**. At the exit line from the charge transfer stage 0 **138** another gate switch **142** enables the transfer of charge from stage 0 **138** to stage 1 **144**. The input switch **136** and the output switch **142** for stage 0 are not in the “on” or closed position at the same time, thus allowing charge to be transferred to and stored at stage 0 **138** prior to being transferred to stage 1 on a subsequent charge transfer cycle. Charge transfer stage **K-1** **144** represents the last charge transfer stage for **K** emitter/detector cycles. Charge is transferred from stage **K-1** **144** to a data bus **146** leading to a 4D frame buffer when the **K-1** output switch **148** is closed. At the end of each of **K** detector integration cycles the grounding switch **149** can be closed to remove any excess charge that may have collected at the photodetector **124**.

(40) FIG. 6 shows electrical timing for the first eleven stages for a 4D camera emitter/detector sequence with **K** stages where **K** equals sixteen. Emitter and detector control circuitry is referenced from the emitter clock **150** that starts at to **152**. An emitter drive pulse **154** activates an emitter for four periods of the emitter clock **150** for each emitter cycle. A detector integration **156** signal determines the integration time for all integration stages in a detector array. Each detector integration period **156** equals six emitter clock **150** periods in this embodiment. The relative timing of the start of the emitter drive pulse **154** and the detector integration **156** will determine the distance range at which each emitter detector cycle will detect objects in a scene. For stage 0 of **K** stages the offset between the emitter start time at to and the start of the detector integration at to results in a minimum time of flight (TOF) of zero emitter clock **150** periods and a maximum TOF of six emitter clock **150** periods, where six periods is determined by the start time of the first emitter pulse **158** and the completion time of the first detector integration cycle **160**. For the second emitter/detector stage of the sequence the start **164** of the detector integration **156** is one emitter clock **150** period after the start **162** of the second emitter drive pulse **154** and the end **166** of the second detector integration **156** is seven emitter clock **150** periods after the start **162** of the emitter drive pulse **154**. The relative timing for the emitter and detector signals for the second emitter/detector stage correspond to a TOF range of one to seven emitter clock **150** periods. By increasing the relative start times for subsequent detector integration **156** periods for each of **K** stages the individual TOF ranges accumulate to define the range for the entire 4D camera capture sequence.

(41) At the completion of the first **160** detector integration **156** period the integrated charge is transferred **168** from each of the **M**×**N** integration elements to each of the **M**×**N** charge transfer stage 0 elements. After the second detector integration **156** period is complete a second charge transfer **170** operation is performed that transfers charge from stage 0 to stage 1 and transfers

charge from the integration stage to charge transfer stage 0. The detector input **172** signal shows times at which charge is being collected at integration stages for the $M \times N$ integration elements. (42) FIG. 7 shows a data element chart **180** that corresponds to the K-stage timing sequence from FIG. 6 where $K=16$, the emitter clock frequency is 80 MHz, and the emitter clock period is 12.5 nSec. The first column **182** shows the stage number of the sixteen stages. Each emitter cycle is four periods with the start times **184** and the end times **186** denoted by their representative $t_{sub.x}$ times. Each detector integration cycle is six periods with the start times **188** and the end times **190** denoted by their representative $t_{sub.x}$ times. The TOF min **192** and TOF max **194** times, denoted by the number of $t_{sub.x}$ periods that define the minimum **192** and maximum **194** times are shown to establish the TOF ranges for each of the K stages. The emitter clock period of 12.5 nSec is used to convert the TOF min. ($t_{sub.x}$) **192** period to a TOF min. (nSec) **196** time and to convert the TOF max. ($t_{sub.x}$) **194** period to a TOF max. (nSec) **198** time. Using the formula for TOF and the speed of light, the distances corresponding to each TOF can be determined by:

Distance=(TOF*c)/2 Eq. 1 Where TOF=time of flight c=speed of light in the medium

(43) Using $c=0.3$ m/nSec as the speed of light the Minimum Dist. (m) **200** and Maximum Dist. (m) **202** values are established for the lower and upper bounds for the range detected for each of the K stages in a 4D camera capture sequence. The intensity (Hex) **204** column shows the digital hexadecimal value for the integrated intensity value for each of the K stages. It is noted that each of the $M \times N$ elements in the detector array will have K intensity values corresponding to the integrated intensities of the K stages. The timing parameters and the TOF values from FIG. 7 are the same for all $M \times N$ elements while the intensity values are only valid for element m,n in the $M \times N$ detector array.

(44) A review of the intensity values **204** shows a minimum value **206** of 0x28 and a maximum value **208** of 0xF0. These values are designated $I_{sub.min}[m,n]=0x28$ and $I_{sub.max}[m,n]=0xF0$. For embodiments that utilize constant pulse width timing for all detector/emitter stages in a capture sequence, the intensity value inserted in the color plane buffer is determined by:

$I_{sub.color}[m,n]=I_{sub.max}[m,n]-I_{sub.min}[m,n]$ Eq. 2

(45) By utilizing Eq. 2 for color plane intensity the effects of ambient light are eliminated by subtracting out the photonic component of intensity $I_{sub.min}[m,n]$ that is due to ambient light on the scene or object. Eq. 2 is an effective approach for eliminating ambient light when the photodetector integration response has a linear relationship to the number of incident photons at the photodetector. For non-linear photonic/charge collection relationships Eq. 2 would be modified to account for the second-order or N-order relationship between incident photons and integrated charge intensity.

(46) For each photodetector m,n in embodiments that utilize multi-color filter elements the $I_{sub.color}[m,n]$ value is stored at location m,n in the color plane that corresponds to the color of the filter. As an example, an embodiment with a Bayer filter pattern (RGBG) will have $M \times N/2$ green filter detectors, $M \times N/4$ blue filter detectors, and $M \times N/4$ red filter detectors. At the completion of K integration stages, subsequent filling of the 4D frame buffer, and determination of the $M \times N$ color values the controller will store the $M \times N/4$ $I_{sub.red}[m,n]$ values at the appropriate locations in the red color plane memory. In turn the controller will determine and store the $M \times N/4$ blue values in the appropriate locations in the blue color plane memory and the $M \times N/2$ green values in the appropriate locations in the green color plane memory.

(47) Referring again to FIG. 7 the maximum value **208** occurs in three consecutive integration cycles, indicating that an entire reflected signal pulse was captured with three consecutive integration windows. By utilizing the minimum distance of the last of the maximum intensity stages the controller determines that the distance to the object for this pixel m,n is 30 meters. The value corresponding to 30 meters is stored at location m,n in the depth map memory. In practice the maximum intensity value may only occur during a single integration cycle. For these cases the controller can utilize partial integration signals to determine object distances. In embodiments the

emitter pulse width and the detector pulse width are selected such that there will be at least one full intensity integration cycle that includes a full-width reflected signal. Furthermore, the relative offset of the start times for successive emitter/detector cycles will ensure that at least one integration cycle will include a leading-edge-clipped reflected signal and at least one integration cycle will include a trailing-edge-clipped reflected signal. The TOF for the leading-edge-clipped and trailing-edge-clipped integration cycles are determined by:

$$(48) \text{ TOF}_{\text{leading-edge-clipped}}(i, m, n) = \text{TOF}_{\min}(i) - \frac{I_{\max}(m, n) - I(i, m, n)}{I_{\max}(m, n) - I_{\min}(m, n)} \quad (\text{Eq. 3})$$

$$\text{TOF}_{\text{trailing-edge-clipped}}(j, m, n) = \text{TOF}_{\max}(j) - \frac{I(j, m, n) - I_{\min}(m, n)}{I_{\max}(m, n) - I_{\min}(m, n)} \quad (\text{Eq. 4})$$

where i is the stage at which the leading-edge-clipped signal is detected j is the stage at which the trailing-edge-clipped signal is detected $I(i, m, n)$ is the intensity value for pixel m, n at stage i $I(j, m, n)$ is the intensity value for pixel m, n at stage j $I_{\text{sub.min}}(m, n)$ is the minimum intensity value for the current emitter/detector sequence $I_{\text{sub.max}}(m, n)$ is the maximum intensity value for the current emitter/detector sequence $\text{TOF}_{\text{sub.min}}(i)$ is the minimum TOF for stage i of the detector sequence $\text{TOF}_{\text{sub.max}}(j)$ is the maximum TOF for stage j of the detector sequence

(49) The embodiments from FIGS. 6 and 7 implement a K-cycle sequence of emitter and detector pulses with an elapsed time of 3.075 microseconds (246 emitter clock periods times 12.5 nSec/emitter clock period). The elapsed time for K emitter/detector cycles that comprise a 4D image cycle is an important parameter for in motion applications. The objective of embodiments for in-motion imaging is to limit the relative movement between the scene and the detectors to 0.05 pixels from the start of the first emitter cycle to the end of the last detector cycle. The relative movement between scene objects and detector pixels is a function of parameters including, but not limited to: camera velocity and direction camera transport velocity and direction object velocity and direction field of view of camera resolution of detector array minimum range of 4D measurements maximum range of 4D measurements

(50) For ground-based vehicle-mounted and low-altitude aircraft-mounted applications the 4D imaging cycle time should be no longer than 50 microseconds. For higher-altitude aircraft at higher speeds the 4D imaging cycle time should be no longer than 10 microseconds. One skilled in the art can envision embodiments where the relative movement between scene objects and the camera exceeds 0.05 pixels. These longer-image-cycle-time embodiments will utilize inter-sample trajectory techniques to account for information from subsequent emitter/detector stages that do not align within the structure of the detector array grid.

(51) Embodiments described in FIGS. 6 and 7 represent signal timing and integrated intensity values for uniform pulse widths in a relatively obstruction-free environment. Sensors will often operate in environments that present less-than-ideal sensing conditions. For optical sensors, conditions that cause incident or reflected signal attenuation require additional consideration when determining color and intensity values and when determining object distances. Atmospheric conditions like dust, rain, fog, sleet and snow will cause signal attenuation due to photon obstruction.

(52) An embodiment in FIG. 8 utilizes the signal timing from FIGS. 6 and 7. The embodiment, in contrast to FIGS. 6 and 7, utilizes RGB information for each pixel. The RGB information may be produced by a Bayer pattern or some other RGB micro-filter configuration at the detector array or it is produced by a monochrome or other grayscale detector array with alternating red, green and blue illumination. FIG. 8 shows a vehicle 300 traversing a roadway in atmospheric conditions that include fog 302. The optical path 304 from the camera 306 to the object 308 is shown projecting through the fog 302. The dotted arcs show the approximate location of the maximum distance for each of the first ten stages of the K stages where $K=16$ in the emitter/detector camera 306 sequence. The stage 0 maximum distance 310 is the closest distance range to the camera 306 and the stage 9 maximum distance 312 is the range that includes the object 308.

(53) FIG. 9 shows the data element chart 314 that corresponds to the K-stage timing sequence where K=16, the emitter clock frequency is 80 MHz, and the emitter clock period is 12.5 nSec. The integrated intensity values for each stage are shown for blue 316, green 318 and red 320 intensities. Since the camera is in a signal-attenuating environment the minimum intensity value for each color is determined utilizing intensity values from distance ranges beyond the detected object. Utilizing the intensity values from stage 15 the values of $I_{sub,min,red}(m,n)=0x26$ 322, $I_{sub,min,green}(m,n)=0x27$ 324, $I_{sub,min,blue}(m,n)=0x28$ 326 are assigned for the present sequence.

(54) Based on the selection of emitter and detector pulse widths for this embodiment, the control algorithm establishes that the intensity values transition from environmental values at stage 6 to ambient values at stage 12. Furthermore, the control algorithm determines that anywhere from one to three stages will contain an integrated signal that includes 100% of the object-reflected waveform. From the data in FIG. 9 the algorithm establishes stage 9 as the stage that contains the full duration of the reflected object waveform. The RGB intensity values for stage 11 328 are higher in amplitude than the values captured during stages 12 through 15. The higher amplitude signifies that the stage 11 328 values are a combination of ambient signals and reflected signals from the object. Furthermore, due to the selected timing of the emitter and detector pulse widths, the reflected intensity value is a partial signal value resulting from a trailing-edge-clipped return signal. The stage 7 332 values are a combination of environment signals and reflected signals from the object, where environment signals are defined as return signals from the environment between the detected object and the sensor. Due to the selected timing of the emitter and detector pulse widths, the reflected intensity value is a partial signal value resulting from a leading-edge-clipped return signal. Utilizing the select signal timing and the sensed intensity values, the values obtained for the select stages are utilized to establish the distance to the object and to determine the color values for the object.

(55) TABLE-US-00001 Object Ambient Stage # Environment Signal % Signal % Signal % 6
 100 0 0 7 E.sub.0 1-E.sub.0 0 8 not used for this computation 9 E.sub.1 1-E.sub.1-A.sub.1
 A.sub.1 10 not used for this computation 11 0 1-A.sub.0 A.sub.0 12 0 0 100

$$I(clr,m,n,s-3)=I_{sub,env}(clr,m,n) \quad \text{Eq. 5}$$

$$I(clr,m,n,s-2)=E_{sub,0}*I_{sub,env}(clr,m,n)+(1-E_{sub,0})I_{sub,obj}(clr,m,n) \quad \text{Eq. 6}$$

$$I(clr,m,n,s+2)=E_{sub,1}*I_{sub,env}(clr,m,n)+(1-E_{sub,1}-A_{sub,1})I_{sub,obj}(clr,m,n)+A_{sub,1}*I_{sub,amb}(clr,m,n) \quad \text{Eq. 7}$$

$$I(clr,m,n,s+2)=A_{sub,0}*I_{sub,amb}(clr,m,n)+(1-A_{sub,0})I_{sub,obj}(clr,m,n) \quad \text{Eq. 8}$$

$$I(clr,m,n,s+3)=I_{sub,amb}(clr,m,n) \quad \text{Eq. 9}$$

$$E_{sub,0}=E_{sub,1}+(2*t_{sub,emitter-clock-cycle})/D \quad \text{Eq. 10}$$

$$A_{sub,1}=A_{sub,0}+(2*t_{sub,emitter-clock-cycle})/D \quad \text{Eq. 11}$$

Where s is the stage number identifier for the detector stage with a 100% reflected signal clr is the color of the pixel m,n is the identifier of the pixel in the array $I_{sub,env}()$ is the detected intensity for the stage with a 100% environmental signal $I_{sub,amb}()$ is the detected intensity for the stage with a 100% ambient signal $I_{sub,obj}()$ is the computed intensity for the object $E_{sub,0}$ is the percentage of the stage s-2 intensity that is due to the environmental signal $E_{sub,1}$ is the percentage of the stage s intensity that is due to the environmental signal $A_{sub,0}$ is the percentage of the stage s intensity that is due to the ambient signal $A_{sub,1}$ is the percentage of the stage s+2 intensity that is due to the ambient signal $t_{sub,emitter-clock-cycle}$ is the period of the emitter clock D is the duty cycle of the emitter/detector for the stage of the sequence and is defined as the emitter-pulse-width divided by the detector-pulse-width for the stage

(56) Utilizing the five equations (Eqs. 6, 7, 8, 10 and 11) with five unknowns, ($I_{sub,obj}()$, $E_{sub,0}$, $E_{sub,1}$, $A_{sub,0}$ and $A_{sub,1}$) the control algorithms determine $I_{sub,obj}()$ for each color and each pixel and assigns the computed intensity values to the appropriate locations in the color frame buffers. The distance to the object is determined by computing TOF to the object based on Eq. 7:

$TOF(clr,m,n)=TOF.sub.min(clr,m,n,s)+E.sub.1*t.sub.detector-pulse-width(s)$ Eq. 12 Where
 $TOF()$ is the time of flight for a particular pixel $TOF.sub.min(s)$ is the minimum time of flight for stage s $E.sub.1$ is the percentage of the stage s intensity that is due to the environmental signal
 $t.sub.detector-pulse-width(s)$ is the width of the detector pulse for stage s

(57) The identification of stage s for Eqs. 5-12 depends on knowledge of the emitter pulse width for each stage and the detector pulse width for each stage. The known pulse widths determine the duty cycle and determine how many stages are involved in the transition from environmental signals to ambient signals for each pixel. Eqs. 5-12 are applicable for embodiments where the emitter pulses are shorter in duration than the detector pulses. For embodiments where emitter pulses are longer than detector pulses Eq. 7 will compute to either $E.sub.1$ or $A.sub.1$ being equal to zero. As a result, two more equations with two more unknowns are necessary to resolve the intensity values of the object. The first additional equation will describe two new unknowns ($A.sub.2$ and $E.sub.2$) as a function of the measured stage intensity and the second additional equation will describe $A.sub.2$ and $E.sub.2$ as a function of the stage duty cycle.

(58) In various embodiments, it will be appreciated that utilizing techniques for evaluating signal attenuation may utilize a minimum of five emitter/detector cycles—one cycle in which an environmental detected signal is determined, one cycle containing a leading edge detected signal of the active pulsed signal, one full-emitter-cycle of the detected signal of the active pulsed signal, one cycle containing a trailing-edge detected signal, and one cycle containing an ambient detected signal. Depending upon timing, field of vision, distances, ambient and environmental conditions, additional emitter/detector cycles may be needed to obtain the necessary information to utilize the techniques for evaluating signal attenuation as described with respect to these embodiments.

(59) For uniform emitter pulses Eq. 3 and Eq. 4 will produce the same value for TOF for each pixel m,n . Due to signal noise and ambient light TOF values based on higher integrated intensity values will produce higher accuracy distance computations than lower integrated intensity values. In embodiments the controller will utilize only one of values from Eq. 3 or Eq. 4 to establish the TOF for the pixel, with the preferred TOF value being selected from the equation that utilizes the largest amplitude integrated intensity value.

(60) Objects farther from 4D cameras will receive less light from emitters than objects closer to the camera. As a result, reflected signals from far objects will have lower intensity than reflected signals from closer objects. One method to compensate for lower intensity return signals is to increase the emitter pulse width and to increase the detector integration time, thus increasing the intensity of the integrated signal for a given object distance. FIG. 10 shows electrical timing for an embodiment that utilizes ten stages in an emitter/detector sequence. In contrast to a prior embodiment wherein all emitter pulses were a constant length and all detector integration cycles were a constant length, this embodiment utilizes steadily increasing emitter pulse periods and detector integration periods.

(61) Stage 0 has a four-period emitter cycle **210** and a six-period detector integration cycle **212**. Stage 1 has a five-period emitter cycle **214** and has a seven-period detector integration cycle **216**. Stage 9 is a special cycle that has a very long emitter pulse **218** and a correspondingly long detector integration cycle **220**. This special long emitter/detector cycle may not be used for distance determination but is used to establish accurate color values for objects that are not very retroreflective at the wavelengths of the camera emitter.

(62) FIG. 11 shows a data table **230** with TOF ranges that correspond to the timing sequences from FIG. 10. Stages 0 through 8 are used to determine values for the depth map for each of the $M \times N$ pixels. Stage 9 is utilized for determining the $I.sub.color[m,n]$ value for the separate color planes.

(63) In previous embodiments the distances to objects computed via TOF were dependent on distance ranges established by the multi-period detector integration cycles. It may be desirable to achieve greater precision for TOF distance measurements. FIG. 12a shows a possible intensity curve **240** for a 50 nSec emitter pulse. The curve **240** has a slower turn-on **242** time ($t.sub.ON$) than

a turn-off **244** time (t.sub.OFF) and has a maximum value **246**. The maximum intensity of an integrated detector value is proportional to the area under the curve **240**. FIG. **12b** **248** shows the cumulative integrated intensity vs. time for the same emitter pulse. For purposes of Eq. 13 and Eq. 14 below the information from FIG. **12b** will be expressed as f(t) where the cumulative intensity value, expressed as a percentage of the maximum intensity value, is a function of time. For partial intensity values in the K-stage sequence, utilizing the cumulative intensity curve, the TOF for the leading-edge-clipped and trailing-edge-clipped integration cycles are determined by:

$$(64) \text{ TOF}_{\text{leading-edge-clipped}}(i, m, n) = \text{TOF}_{\text{min}}(i) - f^{-1} \left\{ \frac{I_{\text{max}}(m, n) - I(i, m, n)}{I_{\text{max}}(m, n) - I_{\text{min}}(m, n)} \right\} \quad (\text{Eq. 13})$$

$\text{TOF}_{\text{trailing-edge-clipped}}(j, m, n) = \text{TOF}_{\text{max}}(j) - f^{-1} \left\{ \frac{I(j, m, n) - I_{\text{min}}(m, n)}{I_{\text{max}}(m, n) - I_{\text{min}}(m, n)} \right\}$ (Eq. 14) where i is the stage at which the leading-edge-clipped signal is detected j is the stage at which the trailing-edge-clipped signal is detected I(i,m,n) is the intensity value for pixel m,n at stage i I(j,m,n) is the intensity value for pixel m,n at stage j I.sub.min(m,n) is the minimum intensity value for the current emitter/detector sequence I.sub.max(m,n) is the maximum intensity value for the current emitter/detector sequence TOF.sub.min(i) is the minimum TOF for stage i of the detector sequence TOF.sub.max(j) is the maximum TOF for stage j of the detector sequence f.sup.-1(t) is the inverse function of f(t) and expresses the point in time during the reflected pulse integration at which the cumulative intensity is equal to the non-integrated portion of the leading edge signal or at which time the cumulative intensity is equal to the integrated portion of the trailing edge signal

(65) In practice f(t) will likely be a non-linear or higher-order relationship between cumulative intensity and time. As such, the inverse function f.sup.-1(t) may be implemented in embodiments as a lookup table or some other numerical conversion function.

(66) FIG. **13a** shows the spectral output **250** for an ideal white (full visible spectrum) emitter. The output energy is at full intensity **252** throughout the range of 400 nanometers to 700 nanometers. FIG. **13b** shows the spectral output **254** for an LED-based white emitter. The curve shows a peak intensity **256** around 430 nanometers, somewhat lower intensity **258** around 520 nanometers, and substantially lower intensity **260** around 660 nanometers. Although not a requirement, in embodiments it is desirable to have a more-uniform spectral output for the 4D camera light source. FIG. **14a** shows the spectral output **262** for three LED types. White **264**, green **266**, and red **268** response curves are shown with their intensity values normalized to one another. The three color component LEDs are combined to produce a cumulative emitter response that is more uniform throughout the desired spectral range of the device. The act of “combining” as described herein refers to selectively populating the emitter array with emitter components that, when switched on simultaneously, will produced the desired cumulative effect for the output spectral response. FIG. **14b** shows the spectral output **270** of the combined signal from the three LED emitter responses depicted in FIG. **14a**.

(67) In embodiments the intensity determination for separate color planes is achieved with an unfiltered detector array and selective use of multi-colored emitters. FIG. **15** shows the spectral output for LEDs utilized in an RGB configuration. Information is produced for the blue color plane when emitter/detector cycles utilize the blue **272** LEDs, information is produced for the green color plane when emitter/detector cycles utilize the green **274** LEDs, and information is produced for the red color plane when emitter/detector cycles utilize the red **276** LEDs. Embodiments that utilize an unfiltered detector array or a wide-bandpass-filtered array and emitters from separate wavelength bands can arrange emitter/detector stages as single-wavelength emitter-detector stages or multi-wavelength emitter/detector stages. An example in Table 1 below shows multiple emitter/detector stages for a K-cycle sequence with K=12, whereby each emitter wavelength is utilized in a round-robin fashion.

(68) TABLE-US-00002 TABLE 1 Round-robin emitters within a single K-stage sequence Stage # Emitter(s) Emitter/Detector Offset 0 Red 0 1 Green 0 2 Blue 0 3 Red 1 4 Green 1 5 Blue 1 6 Red 2

7 Green 2 8 Blue 2 9 Red 3 10 Green 3 11 Blue 3

(69) An example in Table 2 below shows multiple emitter/detector stages for a K-stage sequence with K=12, whereby each emitter wavelength is utilized for K/3 sequential stages.

(70) TABLE-US-00003 TABLE 2 Sequential emitter events within a single K-stage sequence Stage # Emitter(s) Emitter/Detector Offset 0 Red 0 1 Red 1 2 Red 2 3 Red 3 4 Green 0 5 Green 1 6 Green 2 7 Green 3 8 Blue 0 9 Blue 1 10 Blue 2 11 Blue 3

(71) A K-stage sequence with K=12 can also be allocated to a single wavelength emitter, with subsequent K-stage sequences allocated to other wavelengths in a round-robin fashion as shown in Table 3 below.

(72) TABLE-US-00004 TABLE 3 Sequential emitter events in separate K-stage sequences Cycle # Event # Emitter(s) Emitter/Detector Offset 0 0 Red 0 0 1 Red 1 0 2 Red 2 0 3 Red 3 0 4 Red 4 0 5 Red 5 0 6 Red 6 0 7 Red 7 0 8 Red 8 0 9 Red 9 0 10 Red 10 0 11 Red 11 1 0 Green 0 1 1 Green 1 1 2 Green 2 1 3 Green 3 1 4 Green 4 1 5 Green 5 1 6 Green 6 1 7 Green 7 1 8 Green 8 1 9 Green 9 1 10 Green 10 1 11 Green 11 2 0 Blue 0 2 1 Blue 1 2 2 Blue 2 2 3 Blue 3 2 4 Blue 4 2 5 Blue 5 2 6 Blue 6 2 7 Blue 7 2 8 Blue 8 2 9 Blue 9 2 10 Blue 10 2 11 Blue 11

(73) Embodiments that utilize individual detector filters will have certain advantages and disadvantages over embodiments that utilize separate wavelength emitters to achieve multi-color detected signals. Table 4 below compares the relative advantages of embodiments.

(74) TABLE-US-00005 TABLE 4 Comparison of detector filter techniques for visible spectrum embodiments Det. Filter(s) Emitter(s) # of Rs # of Gs # of Bs Advantage RGBG 400-700 nm MxN/4 MxN/2 MxN/4 Inc. range/precision None RGB MxN MxN MxN Inc. spatial resolution

(75) FIG. 16 depicts an embodiment wherein the vehicle headlamps 282, 284 perform the illumination function for the camera 280. Emitter (headlamp) 282, 284 and detector control can reside in the headlamp control circuitry, in the camera 280, in the electronic control module (ECM), or in another on-board location that can ensure controllable timing between emitter events and detector events.

(76) FIG. 17 shows electrical timing elements for a distributed camera system that utilizes vehicle headlamps for emitter illumination. The headlamps are controlled by the ECM during non-imaging periods and by the camera during imaging periods. The ECM Control 286 signal specifies times at which the ECM will control the on and off states of the headlamps and the Camera Control 288 signal specifies when the camera will control the emitters (headlamps). Emitter Drive Pulses 290 and Detector Integration 292 are shown occurring only during the times that Camera Control 288 is in charge of the headlamps. The headlamp output 294 shows the headlamps in multiple on/off states during camera control 288 and in the on state during ECM control 286.

(77) In-motion imaging applications have the advantage of imaging an object from multiple viewpoints and, more importantly, multiple angles. Physical objects possess light-reflecting characteristics that, when sensed properly, can be utilized to categorize objects and even uniquely identify objects and their surface characteristics. FIG. 18 shows a vehicle 340 with a camera 342 on a roadway with an object 344 within the field of view of the camera 342. The location of the vehicle 340 is specified at T0 since it is the initial location of the vehicle 340 when the first image containing the object 344 is captured by the camera 342. Subsequent locations of the vehicle 340 are shown at T1 346, T2 348 and T3 350. The object 344 has a surface that is differentiable from other objects such that the same point on the surface of the object 344 can be identified and analyzed in images obtained from different locations and orientations relative to the object 344.

(78) FIG. 19 shows a flow chart of the steps in the camera processing algorithm to obtain multi-angle information for an object point in the environment. A loop counter is utilized for each object that is processed and analyzed. The loop counter value n is initialized by assigning n=0 360 at the start of each object process. Image analysis software identifies a point P0 on the surface of an object 362 that will be analyzed for possible use in identifying the object using angular intensity profile analysis. Since point P0 is on the surface of an object the algorithm will compute the normal

vector $\theta.\text{sub}.N(n)$ for the surface **364** at point P_0 . Having established the normal vector, the processing algorithm will utilize scene geometry to compute $\theta.\text{sub}.R(n)$, the relative angle **366** between the optical path of the sensor and the normal vector for the object at point P_0 . The angular intensity information for P_0 is stored in the angular response profile memory **368** as a data element of the form $I.\text{sub}.color(n), d(n), \theta.\text{sub}.R(n)$, which specifies the intensity, distance and relative angle for sample n for point P_0 . For embodiments that utilize multiple colors for the detector filter array and/or emitter array each sample n will have a separate response profile memory entry for each color.

(79) Upon completion of the processing for $n=0$ the processing algorithm obtains the next image **370** in a sequence. The image is analyzed to determine if point P_0 is present **372** in the image. If P_0 is present the loop counter is incremented **374** and the algorithm proceeds to the normal vector determination step **364**. If P_0 is not present the algorithm establishes whether there are enough points **376** to identify the object based on angular intensity characteristics. If the minimum requirements are not met the algorithm concludes **384** without identifying the object. If the minimum requirements are met the algorithm creates a plot in 3D space **378** for each color for the intensity information determined for all of the n points. The algorithm will define the object by comparing the collected angular intensity profile to reference characteristic profiles that are stored in a library. The characteristic profiles are retrieved from the library **380** and a correlation is determined **382** for each characteristic profile and the P_0 profile. The characteristic profile with the highest correlation to P_0 is used to determine the object type, class or feature for the object represented by P_0 .

(80) The algorithm from FIG. **19** discusses a single point P_0 . For real-time object analysis the algorithm will typically be running simultaneously for multiple points P_x in an image stream. Results from characteristic angular intensity profile analysis are typically passed to an upstream application for further processing and/or analysis or are packaged in an output data stream with image or depth map information to be sent to another device or application. The algorithm in FIG. **19** utilizes a threshold test to determine if sufficient information is collected to identify the object. Other tests are available and are utilized based on the information in the characteristic profiles. For example, a completeness check can analyze the range of $\theta.\text{sub}.R(n)$ values for P_0 . If the range of values is too narrow the algorithm may discard P_0 due to insufficient information to uniquely characterize the object. In other cases values of $\theta.\text{sub}.R(n)$ for P_0 may be too large and will not correlate to characteristic profiles. Including a $\theta.\text{sub}.R(n)$ amplitude check as part of the threshold check allows the algorithm to discard profiles that will not yield reliable results in the characterization profile correlation.

(81) In practice the library of characteristic angular intensity profiles will contain hundreds or possibly thousands of profiles. Performing correlations on all profiles in real-time is a computationally intensive operation. As a way of parsing the challenge to a more manageable size the analysis functionality on the device can perform image analysis to classify detected objects. Once classified, angular intensity profiles from the detected objects can be compared to only the library profiles that are associated with the identified object class. As an example, the image analysis functionality in a vehicle-mounted application can identify roadway surfaces based on characteristics such as coloration, flatness, orientation relative to the direction of travel, etc. Having established that a profile for a point P_0 is classified as a roadway surface point, the algorithm can access only those characteristic profiles from the library that are classified as road surface characteristics. Some road surface characteristic profiles could include, but not be limited to: Asphalt—smoothness rating A Asphalt—smoothness rating B Asphalt—smoothness rating C Asphalt with surface moisture Asphalt with surface ice Concrete—smoothness rating A Concrete—smoothness rating B Concrete—smoothness rating C Concrete with surface moisture Concrete with surface ice

(82) An object like road signs is another profile class that can be separate in the profile library.

Some road sign characteristic profiles could include, but not be limited to: ASTM Type I ASTM Type III ASTM Type IV—manufacturer A ASTM Type IV—manufacturer M ASTM Type IV—manufacturer N ASTM Type VIII—manufacturer A ASTM Type VIII—manufacturer M ASTM Type IX—manufacturer A ASTM Type IX—manufacturer M ASTM Type XI—manufacturer A ASTM Type XI—manufacturer M

(83) The characteristic profile algorithm specifies correlation as the means to compare characteristic profiles and to select the most representative characteristic profile for the object represented by P0. Those reasonably skilled in the art can devise or utilize other methods to select the most representative characteristic profile based on the information collected and analyzed for the object represented by P0.

(84) FIG. 20 shows a view from inside a passenger compartment of a vehicle 400. A steering wheel 402 is shown, although embodiments can be utilized in autonomous vehicles without steering wheels 402. Environmental conditions 414 outside the vehicle 400 produce low visibility due to phenomena like fog, rain, snow, sleet, dust. Alternately, or in addition, the see-through elements like the windshield 404 and the rear window (not shown) may have surface irregularities or coatings that limit the viewing of external conditions 414 from inside the vehicle 400. A front-facing camera 406 is mounted inside the vehicle 400 behind the windshield 404 near the rear-view mirror 408 to provide a forward-imaging view. The front-facing camera 406 control system detects an environmental condition for low visibility and projects 412 the image 410 onto the windshield 404 or other heads-up display feature to allow unobstructed viewing of the objects in the low-visibility environment.

(85) The rear-view mirror 408 displays an unobstructed view from a rear-facing camera (not shown) that is in a rear-facing orientation mounted at the rear of the vehicle 400 or inside the vehicle 400 projecting through the rear window. Environmental 416 obstructions for the side of the vehicle 400 are addressed with features in the side mirror 418. A rear oblique-angle camera 420 detects obstructed environmental conditions 416 and projects an obstruction-free image 422 on the mirror for use by the vehicle 400 operator. Alternately, or in addition, the obstruction-free image 422 is delivered to the vehicle control system for autonomous or semi-autonomous driving systems. An indicator 424 on the side mirror indicates the presence of objects within a certain space, thus assisting the vehicle 400 operator in maneuvers like lane changes.

(86) The other embodiments, the processing system can include various engines, each of which is constructed, programmed, configured, or otherwise adapted, to autonomously carry out a function or set of functions. The term engine as used herein is defined as a real-world device, component, or arrangement of components implemented using hardware, such as by an application specific integrated circuit (ASIC) or field-programmable gate array (FPGA), for example, or as a combination of hardware and software, such as by a microprocessor system and a set of program instructions that adapt the engine to implement the particular functionality, which (while being executed) transform the microprocessor or controller system into a special-purpose device. An engine can also be implemented as a combination of the two, with certain functions facilitated by hardware alone, and other functions facilitated by a combination of hardware and software. In certain implementations, at least a portion, and in some cases, all, of an engine can be executed on the processor(s) of one or more computing platforms that are made up of hardware that execute an operating system, system programs, and/or application programs, while also implementing the engine using multitasking, multithreading, distributed processing where appropriate, or other such techniques.

(87) Accordingly, it will be understood that each processing system can be realized in a variety of physically realizable configurations, and should generally not be limited to any particular implementation exemplified herein, unless such limitations are expressly called out. In addition, a processing system can itself be composed of more than one engine, sub-engines, or sub-processing systems, each of which can be regarded as a processing system in its own right. Moreover, in the

embodiments described herein, each of the various processing systems may correspond to a defined autonomous functionality; however, it should be understood that in other contemplated embodiments, each functionality can be distributed to more than one processing system. Likewise, in other contemplated embodiments, multiple defined functionalities may be implemented by a single processing system that performs those multiple functions, possibly alongside other functions, or distributed differently among a set of processing system than specifically illustrated in the examples herein.

(88) Embodiments utilize high-speed components and circuitry whereby the relative movement of the device and/or scene could be defined as the movement of less than the inter-element spacing in the detector array. For embodiments wherein the relative movement is small the processing software can assume the axis of the 3D volumetric computations is normal to the detector elements in the array. For relative movement greater than the inter-element spacing in the detector array during the timeframe of the emitter cycles the frame buffer analysis software will need to perform 3D analysis of the sampled waveforms whereby the representations have an axis that is non-normal to the detector elements in the array.

(89) The electrical circuitry of embodiments is described utilizing semiconductor nomenclature. In other embodiment circuitry and control logic that utilizes optical computing, quantum computing or similar miniaturized scalable computing platform may be used to perform part or all of the necessary high-speed logic, digital storage, and computing aspects of the systems described herein. The optical emitter elements are described utilizing fabricated semiconductor LED and laser diode nomenclature. In other embodiments the requirements for the various techniques described herein may be accomplished with the use of any controllable photon-emitting elements wherein the output frequency of the emitted photons is known or characterizable, is controllable with logic elements, and is of sufficient switching speed.

(90) In some embodiments, the light energy or light packet is emitted and received as near-collimated, coherent, or wide-angle electromagnetic energy, such as common laser wavelengths of 650 nm, 905 nm or 1550 nm. In some embodiments, the light energy can be in the wavelength ranges of ultraviolet (UV)—100-400 nm, visible—400-700 nm, near infrared (NIR)—700-1400 nm, infrared (IR)—1400-8000 nm, long-wavelength IR (LWIR)—8 μ m-15 μ m, far IR (FIR)—15 μ m-1000 μ m, or terahertz—0.1 mm-1 mm. Various embodiments can provide increased device resolution, higher effective sampling rates and increased device range at these various wavelengths.

(91) Detectors as utilized in the various embodiments refer to discrete devices or a focal plane array of devices that convert optical energy to electrical energy. Detectors as defined herein can take the form of PIN photodiodes, avalanche photodiodes, photodiodes operating at or near Geiger mode biasing, or any other devices that convert optical to electrical energy whereby the electrical output of the device is related to the rate at which target photons are impacting the surface of the detector.

(92) Persons of ordinary skill in the relevant arts will recognize that embodiments may comprise fewer features than illustrated in any individual embodiment described above. The embodiments described herein are not meant to be an exhaustive presentation of the ways in which the various features of the embodiments may be combined. Accordingly, the embodiments are not mutually exclusive combinations of features; rather, embodiments can comprise a combination of different individual features selected from different individual embodiments, as understood by persons of ordinary skill in the art. Moreover, elements described with respect to one embodiment can be implemented in other embodiments even when not described in such embodiments unless otherwise noted. Although a dependent claim may refer in the claims to a specific combination with one or more other claims, other embodiments can also include a combination of the dependent claim with the subject matter of each other dependent claim or a combination of one or more features with other dependent or independent claims. Such combinations are proposed herein unless it is stated that a specific combination is not intended. Furthermore, it is intended also to include features of a claim in any other independent claim even if this claim is not directly made

dependent to the independent claim.

(93) Any incorporation by reference of documents above is limited such that no subject matter is incorporated that is contrary to the explicit disclosure herein. Any incorporation by reference of documents above is further limited such that no claims included in the documents are incorporated by reference herein. Any incorporation by reference of documents above is yet further limited such that any definitions provided in the documents are not incorporated by reference herein unless expressly included herein.

(94) For purposes of interpreting the claims, it is expressly intended that the provisions of Section 112, sixth paragraph of 35 U.S.C. are not to be invoked unless the specific terms “means for” or “step for” are recited in a claim.

Claims

1. A lighting control module for a vehicle having a set of headlamps configured to selectively illuminate a scene proximal to the vehicle and a camera system configured to selectively generate an image of the scene, the lighting control module comprising: a controller configured to generate a control signal having a first state representative of non-imaging periods of the camera system and a second state representative of imaging periods of the camera system; an electronic control module configured to control on/off states of the set of headlamps during the non-imaging periods of the camera system based on the control signal; and a camera control configured to pulse the set of headlamps multiple times per second with a plurality of different predetermined pulse lengths during a set of the imaging periods of the camera system based on the control signal.
2. The lighting control module of claim 1, wherein the set of headlamps comprises at least one light-emitting diode.
3. The lighting control module of claim 1, wherein the set of headlamps comprises a plurality of light-emitting diodes having multiple color components.
4. The lighting control module of claim 1, wherein a quantity of camera control on states during an imaging period of the set of imaging periods is zero, one, or more than one.
5. The lighting control module of claim 1, wherein the set of headlamps includes at least one emitter and the camera system includes an array of detectors, and wherein the camera control is configured to control the multiple on/off states of the set of headlamps during the set of imaging periods of the camera system by: causing the at least one emitter to begin to emit an active sequence of pulses at a first time and to cause the array of detectors to begin to receive light at a second time after the first time in an emitter/detector cycle, wherein a control circuitry is configured to vary an elapsed time between the first time and the second time in K successive emitter/detector cycles; and causing a processing system of the camera system to: generate and store digital information corresponding to the light received by each detector in the array of detectors, the digital information for each detector in the array of detectors being sampled and stored in one of K frame buffers corresponding to one of the K successive emitter/detector cycles, and analyze the digital information for each detector and construct a representation of at least a portion of the scene based at least in part on (i) time-of-flight (TOF) data in the digital information corresponding to the active sequence of pulses received by different ones of the K frame buffers and (ii) relative timing differences of the sequence of pulses for each of the K successive emitter/detector cycles.
6. The lighting control module of claim 5, wherein a wavelength range defined for the at least one emitter is selected from the group consisting of (i) 100 nanometers (nm) to 400 nm, (ii) 400 nm to 700 nm, (iii) 700 nm to 1400 nm, (iv) 1400 nm to 8000 nm, (v) 8 micrometers (micron) to 15 micron, (vi) and 15 micron to 1000 micron.
7. The lighting control module of claim 1, wherein the control signal is an electrical control signal.
8. The lighting control module of claim 1, wherein imaging periods are equal to or less than 100

microseconds.

9. A vehicle comprising: a set of headlamps configured to selectively illuminate a scene proximal to the vehicle; a camera system configured to selectively capture an image of the scene; and a lighting control module comprising: a controller configured to generate a control signal having a first state representative of non-imaging periods of the camera system and a second state representative of imaging periods of the camera system; an electronic control module configured to control on/off states of the set of headlamps during the non-imaging periods of the camera system based on the control signal; and a camera control configured to pulse the set of headlamps multiple times per second during the imaging periods of the camera system based on the control signal.

10. The vehicle of claim 9, wherein the set of headlamps comprises at least one light-emitting diode.

11. The vehicle of claim 9, wherein the set of headlamps comprises a plurality of light-emitting diodes having multiple color components.

12. The vehicle of claim 9, wherein a quantity of camera control on states during an imaging period of the set of imaging periods is zero, one, or more than one.

13. The vehicle of claim 9, wherein the set of headlamps includes at least one emitter and the camera system includes an array of detectors, and wherein the camera control is configured to control the multiple on/off states of the set of headlamps during the set of imaging periods of the camera system by: causing the at least one emitter to begin to emit an active sequence of pulses at a first time and to cause the array of detectors to begin to receive light at a second time after the first time in an emitter/detector cycle, wherein a control circuitry is configured to vary an elapsed time between the first time and the second time in K successive emitter/detector cycles; and causing a processing system of the camera system to: generate and store digital information corresponding to the light received by each detector in the array of detectors, the digital information for each detector in the array of detectors being sampled and stored in one of K frame buffers corresponding to one of the K successive emitter/detector cycles, and analyze the digital information for each detector and construct a representation of at least a portion of the scene based at least in part on (i) time-of-flight (TOF) data in the digital information corresponding to the active sequence of pulses received by different ones of the K frame buffers and (ii) relative timing differences of the sequence of pulses for each of the K successive emitter/detector cycles.

14. The vehicle of claim 13, wherein a wavelength range defined for the at least one emitter is selected from the group consisting of (i) 100 nanometers (nm) to 400 nm, (ii) 400 nm to 700 nm, (iii) 700 nm to 1400 nm, (iv) 1400 nm to 8000 nm, (v) 8 micrometers (micron) to 15 micron, (vi) and 15 micron to 1000 micron.

15. The vehicle of claim 9, wherein the control signal is an electrical signal.

16. A method of vehicle and camera lighting control, comprising: generating a control signal having a first state representative of non-imaging periods of a camera system and a second state representative of imaging periods of the camera system; determining a non-imaging period based on the control signal; controlling the on/off state of a set of headlamps during the determined non-imaging period; determining an imaging period based on the control signal; and pulsing the set of headlamps multiple times per second during the determined imaging period.

17. The method of claim 16, further comprising: selectively illuminating, by the headlamps, a scene proximal to a vehicle; and capturing, by the camera system, one or more images of the scene during the determined imaging period.

18. The method of claim 16, further comprising: emitting light at a first wavelength during the determined non-imaging period; and emitting light at a second wavelength, different from the first wavelength, during the determined imaging period.

19. The method of claim 16, further comprising: emitting, by the headlamps, an active sequence of pulses at a first time; receiving, by an array of detectors, light at a second time after the first time in an emitter/detector cycle; generating and storing digital information corresponding to the light

received by each detector in the array of detectors, the digital information for each detector in the array of detectors being sampled and stored in one of K frame buffers corresponding to one of the K successive emitter/detector cycles; and analyzing the digital information for each detector and construct a representation of at least a portion of the scene based at least in part on (i) time-of-flight (TOF) data in the digital information corresponding to the active sequence of pulses received by different ones of the K frame buffers and (ii) relative timing differences of the sequence of pulses for each of the K successive emitter/detector cycles.

20. The method of claim 19, further comprising varying an elapsed time between the first time and the second time in K successive emitter/detector cycles.

21. The method of claim 19, wherein a wavelength range defined for the at least one emitter is selected from the group consisting of (i) 100 nanometers (nm) to 400 nm, (ii) 400 nm to 700 nm, (iii) 700 nm to 1400 nm, (iv) 1400 nm to 8000 nm, (v) 8 micrometers (micron) to 15 micron, (vi) and 15 micron to 1000 micron.

22. The method of claim 16, wherein the control signal is an electrical control signal.
